Computer Vision and Image Understanding 264 (2026) 104650

Contents lists available at ScienceDirect

Computer Vision and Image Understanding

journal homepage: www.elsevier.com/locate/cviu

Indoor UAV navigation using event cameras and intermediate frame reconstruction

David Tejero-Ruiz [a,b], David Solís-Martín [a,*], Francisco J. Pérez-Grau [b], Joaquín Borrego-Díaz [a]

[a] Department of Computer Science and Artificial Intelligence, Seville University, Seville, Spain
[b] Advanced Center for Aerospace Technologies (CATEC), Seville, Spain

ARTICLE INFO

Communicated by Limin Wang

Keywords:
Event camera
Unmanned Aerial Vehicle
Deep learning
Visual-inertial odometry
Sensor fusion
Pose estimation

ABSTRACT

Indoor UAV navigation faces significant challenges due to GPS signal absence and limitations of conventional visual-inertial systems under challenging lighting and motion conditions. This paper presents an event-based visual-inertial odometry system that addresses these limitations through intermediate frame reconstruction from event streams combined with established odometry algorithms. The approach leverages event cameras' unique characteristics — microsecond temporal resolution, high dynamic range (120 dB), and motion blur immunity — to maintain stable navigation performance under conditions that cause conventional systems to fail. The system achieves real-time operation at 30 Hz frame reconstruction and 20 Hz pose estimation on embedded hardware, consuming 15 W power while adding only 50 g to the UAV platform. Experimental validation in controlled indoor environments demonstrates mean absolute pose errors of 26–42 cm across different operational conditions, comparable to conventional visual-inertial systems. Critically, the system maintains stable performance during rapid lighting transitions, showing only 59% performance degradation compared to baseline conditions, while conventional cameras typically experience complete tracking failure. The results establish event-based visual-inertial odometry as a viable alternative for indoor UAV navigation, particularly in applications requiring environmental robustness over marginal accuracy improvements under optimal conditions.

1. Introduction

The effectiveness of robotic navigation systems depends fundamentally on the quality of embedded sensors and the operational environment characteristics. Sensors provide essential data for estimating position and orientation, while environmental conditions determine the reliability and accuracy of these measurements. For Unmanned Aerial Vehicles (UAVs), navigation requirements are particularly demanding due to their dynamic nature and safety-critical applications.

Outdoor UAV navigation achieves sufficient precision for safe operation through established positioning technologies. The combination of Global Navigation Satellite Systems (GNSS), Inertial Measurement Units (IMUs), and magnetometers enables accurate pose estimation under normal conditions, with typical horizontal accuracies of 1–3 meters. Research has demonstrated the effectiveness of these systems in practical applications, including automated delivery services (Patrik et al., 2019). However, indoor navigation presents fundamentally different challenges due to limited or absent GNSS signals, creating what are commonly termed GNSS-denied environments.

UAVs, particularly multi-rotor systems, are increasingly deployed for indoor tasks ranging from surveillance and inventory management to operations in hazardous environments such as infrastructure inspection and search and rescue missions (Shakhatreh et al., 2019). These applications require navigation systems that can operate reliably in complex environments while maintaining real-time performance on resource-constrained platforms.

Current indoor navigation solutions face significant limitations. Visual-inertial odometry systems struggle with motion blur during aggressive maneuvers and fail under rapid lighting changes. LiDAR systems provide accurate depth information but impose weight penalties (100–500 g) and high costs ($5,000+) unsuitable for small UAV platforms. This creates a clear need for lightweight, cost-effective navigation solutions that perform reliably under challenging motion and lighting conditions.

This paper proposes an event camera-based visual-inertial odometry system specifically designed for indoor UAV navigation.

* Corresponding author.
E-mail addresses: dtejero@catec.aero (D. Tejero-Ruiz), dsolis@us.es (D. Solís-Martín), fjperez@catec.aero (F.J. Pérez-Grau), jborrego@us.es (J. Borrego-Díaz).

https://doi.org/10.1016/j.cviu.2026.104650
Received 12 August 2025; Received in revised form 20 October 2025; Accepted 7 January 2026
Available online 10 January 2026
1077-3142/© 2026 The Authors. Published by Elsevier Inc. This is an open access article under the CC BY-NC license (http://creativecommons.org/licenses/by-nc/4.0/).

Event cameras offer distinctive characteristics that address key limitations of conventional sensing: microsecond temporal resolution, high dynamic range, and immunity to motion blur. While modern HDR frame sensors can match an event camera dynamic range (120 dB), they remain frame-based and thus subject to exposure-time trade-offs and motion blur under fast dynamics. Event cameras, instead, provide asynchronous per-pixel readout with microsecond timestamps and no motion blur, which proves useful during abrupt illumination transitions or rapid motion.

1.1. Indoor UAV navigation: Current approaches

Indoor UAV navigation has been addressed through several technological approaches, each presenting distinct trade-offs between accuracy, cost, and operational constraints.

Visual-inertial odometry represents the most widely adopted approach for indoor navigation due to its cost-effectiveness and moderate computational requirements. Established systems include VINS-Mono (Qin et al., 2018), which achieves real-time performance with translational accuracy of 0.1–0.5 m in structured environments, OpenVINS (Geneva et al., 2020), offering improved numerical stability, and HybVIO (Seiskari et al., 2022), demonstrating enhanced performance in dynamic scenarios.

However, conventional camera-based systems face fundamental limitations in UAV applications. Motion blur during aggressive maneuvers (typical in UAV flight) degrades feature tracking quality, leading to pose estimation errors exceeding 1 m over 100 m trajectories (Macario Barros et al., 2022). Rapid lighting changes, common in indoor environments with varying illumination sources, cause feature tracking failures and system divergence (Minoda et al., 2021). Frame-rate limitations (typically 30–60 Hz) create temporal gaps that miss high-frequency motion dynamics essential for UAV control.

LiDAR systems provide accurate environmental reconstruction through point cloud generation. Two-dimensional LiDAR implementations (Sun et al., 2023) achieve centimeter-level accuracy in structured environments but require planar motion assumptions unsuitable for UAV applications. Three-dimensional LiDAR systems (Caballero et al., 2021) overcome these limitations, providing full 6-DOF pose estimation with sub-centimeter accuracy in feature-rich environments.

Despite their accuracy advantages, LiDAR systems present prohibitive constraints for small UAV platforms. Weight requirements (100–500 g for 3D units) significantly reduce flight time, while costs ($5000-$50,000) limit widespread deployment. Power consumption (5–20 W) further constrains battery life, and mechanical scanning components reduce system reliability.

Ultra-wideband (UWB) systems achieve sub-decimeter accuracy but require pre-deployed infrastructure with known anchor positions, limiting applicability to prepared environments. Optical flow sensors provide velocity estimates suitable for hovering applications but lack absolute positioning capability and struggle in texture-poor environments. Magnetic field-based systems suffer from interference in indoor environments with electronic devices and metallic structures.

1.2. Event camera technology and applications

Event cameras represent an emerging sensing paradigm that addresses fundamental limitations of conventional vision systems through asynchronous, pixel-level change detection.

Event cameras operate by detecting logarithmic intensity changes at individual pixels, generating sparse event streams with microsecond temporal resolution. Key advantages include: (1) High dynamic range (120 dB) enabling operation under extreme lighting conditions, (2) Absence of motion blur due to continuous sensing rather than frame-based capture, (3) Low latency response (microseconds vs. milliseconds for conventional cameras), and (4) Reduced data rates in static scenes due to sparse output representation.

These characteristics directly address the primary limitations of conventional cameras in UAV navigation: motion blur during rapid maneuvers and failure under challenging lighting conditions.

Visual feature tracking using event cameras has demonstrated advantages in dynamic scenarios. Early works (Delbruck and Lang, 2013; Glover and Bartolozzi, 2017) adapted classical algorithms to the event paradigm, showing improved tracking performance under motion blur conditions. The asynchronous nature enables continuous feature tracking without frame-rate limitations, crucial for high-speed UAV applications.

Video reconstruction from events bridges event-based sensing with established computer vision algorithms. Classical approaches (Cook et al., 2011) showed limited success until deep learning revolutionized the field. The E2VID architecture (Rebecq et al., 2019) achieved high-quality reconstruction using recurrent convolutional neural networks, while FireNet (Scheerlinck et al., 2020) reduced inference time by 40% while maintaining reconstruction quality. Recent advances include ET-Net (Weng et al., 2021) leveraging transformers and HyperE2VID (Ercan et al., 2024) incorporating hypernetworks for improved adaptability.

Event-based SLAM addresses the dual challenge of environment mapping and pose estimation. Early approaches (Gallego et al., 2018) achieved continuous pose tracking using pre-computed depth maps but required prior environmental knowledge. Advanced sensor fusion approaches integrate event cameras with IMUs (Vidal et al., 2018) and conventional cameras, while probabilistic frameworks (Hidalgo-Carrió et al., 2022) combine frames and events for improved robustness.

Stereo event camera configurations (Zhou et al., 2021) demonstrate improved depth estimation accuracy but increase system complexity and calibration requirements. Current limitations include computational complexity preventing real-time embedded implementation and limited evaluation in realistic UAV scenarios.

1.3. Event cameras in UAV applications

Several studies have explored event cameras specifically for UAV applications, primarily focusing on obstacle avoidance and control tasks.

Obstacle detection and avoidance represents the primary application area. Falanga et al. (2020) demonstrated successful detection and avoidance of fast-moving obstacles using ego-motion compensated event processing, achieving 3.5 ms reaction times compared to 50 ms for conventional cameras. However, this approach focused solely on obstacle avoidance rather than full pose estimation.

Neuromorphic approaches include (Zhang et al., 2023), employing Spiking Neural Networks for dynamic obstacle identification, though requiring additional depth sensors. Vitale et al. (2021) developed a position control system using SNNs on neuromorphic hardware, achieving low latency and power consumption but limited to simple hovering tasks.

1.4. Research gap analysis

The literature review reveals several critical gaps in current event-based navigation systems for UAV applications:

1. **Limited embedded implementations**: Existing event-based SLAM systems require significant computational resources (typically > 10 W GPU processing), preventing deployment on weight and power-constrained UAV platforms where processing budgets are typically <5 W.
2. **Incomplete system integration**: Most works focus on individual components (feature tracking, reconstruction, or mapping) without demonstrating complete navigation systems suitable for autonomous UAV operation.

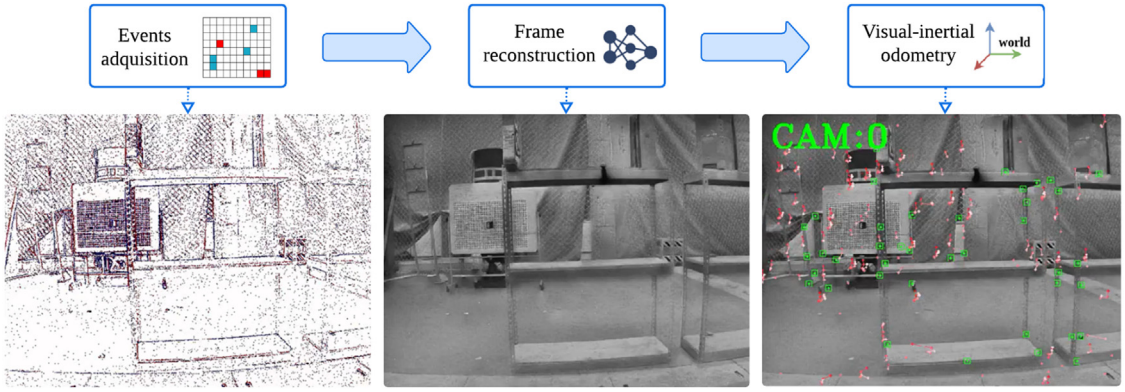


Fig. 1. Proposed event-based visual-inertial odometry system architecture.

3. **Lack of quantitative comparison**: Few studies provide systematic comparison with conventional visual-inertial systems under identical conditions, particularly for the challenging motion and lighting scenarios encountered in UAV navigation.
4. **Missing robustness evaluation**: While event cameras theoretically address lighting and motion challenges, systematic evaluation of these advantages in practical UAV navigation scenarios remains limited.
5. **Real-time constraints**: Existing approaches often prioritize reconstruction quality over computational efficiency, failing to meet the strict real-time requirements (>20Hz) necessary for UAV control loops.

1.5. Proposed approach and contributions

This work addresses the identified research gaps by developing a complete event-based visual-inertial odometry system specifically designed for real-time UAV navigation. Unlike previous approaches that rely on direct event processing or require multiple sensors, our method leverages intermediate frame reconstruction to bridge event sensing with established visual-inertial algorithms.

The main contributions of this work include:

- A complete event-based visual-inertial odometry system integrating monocular event camera and IMU in a lightweight configuration, achieving real-time performance (30 Hz frame reconstruction, 20 Hz pose estimation) on embedded hardware with 15 W power consumption and 50 g weight penalty suitable for small UAV platforms.
- Systematic adaptation and optimization of the FireNet reconstruction architecture for UAV navigation applications, including domain-specific fine-tuning using ESIM-generated synthetic datasets and comprehensive noise analysis with practical hot pixel mitigation strategies.
- Quantitative experimental validation in controlled indoor environments using motion capture ground-truth, demonstrating mean absolute pose errors of 26–42 cm across different operational conditions and establishing performance levels comparable to conventional visual-inertial systems.
- Demonstration of enhanced robustness under challenging conditions, showing stable operation during rapid lighting transitions with only 59% performance degradation compared to baseline, while maintaining consistent performance during actual UAV flight operations with realistic vibrations and dynamics.
- Complete software implementation within the ROS framework with processing latencies of 50-80ms suitable for UAV control loops, including three open-source packages enabling reproducible research and practical deployment of event-based navigation systems.
- Systematic performance analysis identifying specific advantages and limitations of event-based approaches for indoor UAV navigation, providing quantitative benchmarks and implementation insights that inform future research directions in neuromorphic sensing for autonomous aerial systems.

The synthetic frame reconstruction approach, while initially appearing counterintuitive for event cameras, incorporates incremental light variation information at high temporal resolution, enabling high dynamic range reconstruction without motion blur. The lightweight neural network architecture achieves low inference times (10-15ms) with acceptable reconstruction quality, making the system suitable for integration with UAV autopilot systems in closed-loop control applications.

The remainder of this paper is organized as follows: Section 2 presents our system architecture, Section 3 describes experimental validation, Section 4 analyzes the experimental results, the practical implications and discusses limitations of the proposed approach and Section 5 discusses results and future work.

2. Methodology

This section presents the complete system architecture for event-based visual-inertial odometry, including hardware platforms, experimental setup, and algorithmic components. The proposed approach addresses the identified research gap by combining event camera sensing with established visual-inertial odometry techniques through an intermediate frame reconstruction strategy.

2.1. System architecture overview

The proposed localization system integrates event cameras with inertial measurements through a two-stage processing pipeline: (1) deep learning-based frame reconstruction from event streams, and (2) conventional visual-inertial odometry using reconstructed frames and IMU data. This architecture leverages the advantages of event cameras (high dynamic range, motion blur immunity) while maintaining compatibility with established VIO algorithms.

Fig. 1 summarizes the processing pipeline. Raw event streams and IMU measurements serve as inputs to the frame reconstruction module, which generates synthetic images at fixed intervals. These reconstructed frames, combined with inertial data, feed into a visual-inertial odometry algorithm that estimates 6-DOF pose in real-time.

The complete software architecture of the proposed system is presented in Fig. 2, which illustrates the data flow from sensor acquisition to pose estimation. The pipeline begins with the event camera and IMU providing raw measurements at the hardware layer, which are then processed through several stages: preprocessing for noise

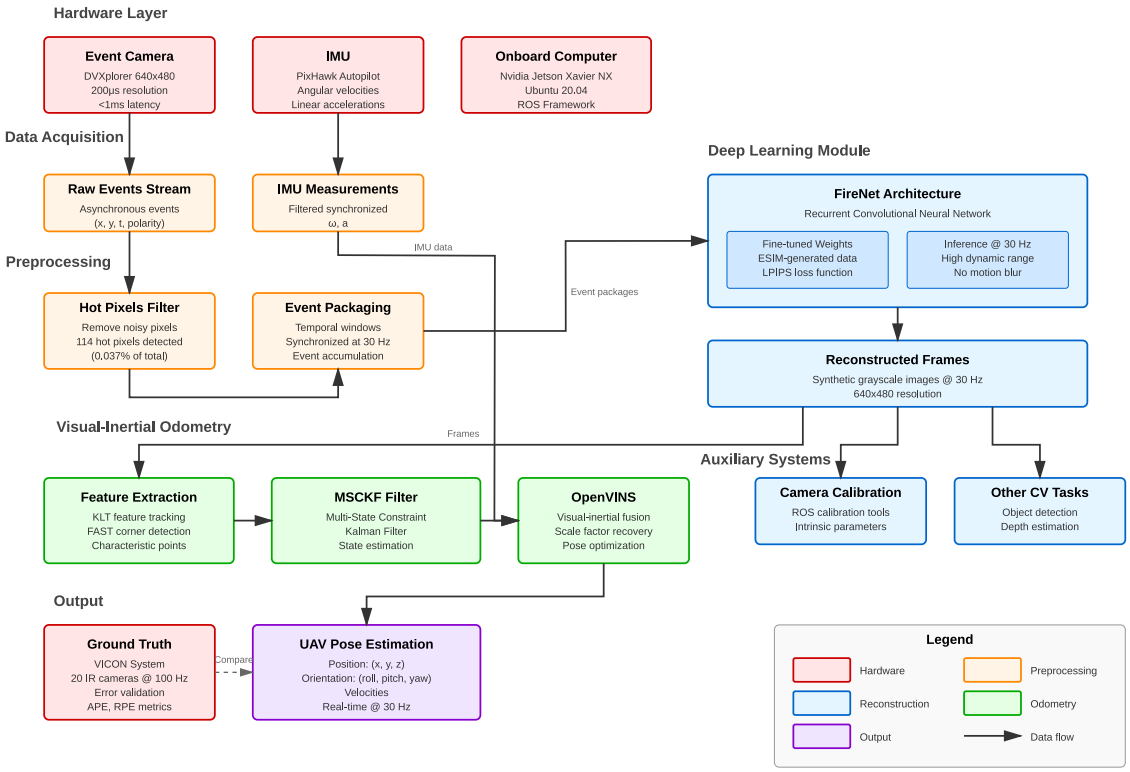


Fig. 2. Systematic block diagram of the proposed localization system. The architecture shows the complete data flow from event camera and IMU sensors through preprocessing, FireNet-based frame reconstruction, and visual-inertial odometry (OpenVINS with MSCKF filter) to obtain real-time UAV pose estimation.

removal and event packaging, deep learning-based frame reconstruction using FireNet, and visual-inertial odometry through OpenVINS with MSCKF filtering. This modular architecture, implemented within the ROS framework on the onboard computer, enables real-time UAV localization at 30 Hz. The following subsections provide a detailed description of each module and its specific implementation.

2.2. Hardware platform

The primary sensor is an IniVation DVXplorer event camera with 640 × 480 pixel resolution and 200 microsecond temporal resolution (see Fig. 3). The camera provides sub-millisecond latency according to manufacturer specifications and includes manual adjustments for focus, zoom, and aperture. The sensor generates asynchronous event streams where each event encodes pixel coordinates, timestamp, and polarity (brightness increase/decrease).

To mitigate high-frequency vibrations during flight, the camera is mounted on the UAV using rubber dampers and a slightly tilted configuration to optimize the field of view for navigation tasks. An infrared filter is installed to prevent interference from the motion capture system's infrared illumination.

The experimental platform consists of a DJI F550 hexarotor frame equipped with a Pixhawk Orange Cube autopilot running ArduPilot firmware (Fig. 4). An NVIDIA Jetson Xavier NX serves as the onboard computer, providing sufficient computational resources for real-time algorithm execution while maintaining acceptable weight and power constraints for the UAV platform.

The system includes wireless communication capabilities for data logging and remote monitoring during flight tests. IMU measurements are obtained directly from the autopilot, ensuring pre-filtered and calibrated inertial data consistent with the UAV's control system.



Fig. 3. DVXplorer event camera.

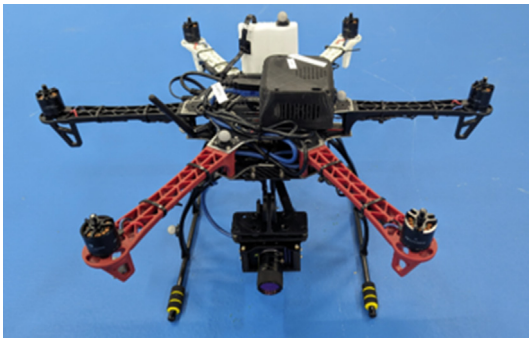


Fig. 4. Experimental UAV platform.



Fig. 5. CATEC multi-UAS laboratory testing environment.



Fig. 6. VICON motion capture camera.

2.3. Experimental environment

Validation experiments are conducted in the CATEC multi-UAS laboratory, a 15 × 15 × 5 meter indoor space equipped with a VICON motion capture system. Twenty VICON cameras operating at 500 Hz provide millimeter-level pose accuracy through infrared marker triangulation, serving as ground-truth for quantitative evaluation (see Fig. 6).

The environment is enhanced with visual features including shelves, objects, and textured surfaces to provide sufficient visual content for feature-based navigation algorithms (Fig. 5). This configuration enables comprehensive evaluation of the proposed system under realistic indoor navigation conditions .

2.4. Event-based frame reconstruction

Frame reconstruction employs the FireNet architecture (Scheerlinck et al., 2020), selected for its balance between reconstruction quality and computational efficiency. FireNet uses a recurrent convolutional neural network that processes temporally accumulated event representations, generating synthetic frames that preserve the high dynamic range and motion blur immunity characteristics of event cameras.

Although the DVXplorer provides events with ∼200 µs timestamp precision, reconstruction is not performed at microsecond rates. Instead, the μs-stamped events are integrated into fixed 33 ms windows to produce 30 Hz frames for the VIO back-end. The microsecond timing is still useful because it reduces within-window motion accumulation (lower effective motion blur in the synthetic frame), improves robustness during abrupt illumination transitions without exposure retuning, and enables accurate interpolation with IMU data. In other words, microsecond-level timestamps are not used to output μs-rate video, but to condition the 30 Hz reconstruction with less aliasing and better temporal alignment under medium-dynamics indoor flight.

The network processes event data accumulated over fixed time windows (33 ms, corresponding to 30 Hz output), generating reconstructed frames with the same resolution as the input event stream. This fixed-rate approach ensures predictable computational load and compatibility with downstream visual-inertial odometry algorithms.

Table 1
Quantitative evaluation of fine-tuning process on synthetic validation data.

Network weights	LPIPS	SSIM	MSE
Original	0.1220	0.6097	0.0617
Fine-tuned	0.1214	0.6102	0.0611

To address the domain gap between generic training data and specific application requirements, we implement a fine-tuning strategy using application-specific synthetic datasets. The ESIM simulator (Rebecq et al., 2018) generates paired event-frame data using calibrated camera parameters matching the DVXplorer specifications.

The fine-tuning dataset consists of simulated camera trajectories in a virtual environment reconstructed from photographs of the actual testing facility (Fig. 7). Random three-dimensional camera movements are simulated over 70-second sequences, generating 20 training sequences of 10 event-frame pairs each. Gaussian noise is added to event packages to improve reconstruction performance under realistic sensor conditions.

Training uses perceptual loss based on VGG-19 features (LPIPS loss), which better captures visual similarity compared to pixel-wise metrics. Table 1 presents quantitative comparison between original and fine-tuned networks, showing modest but consistent improvements across multiple metrics.

The network is trained using the Adam optimizer with a fixed learning rate. To enhance robustness, online augmentation is applied by injecting Gaussian noise into the event packages during data loading. This domain adaptation process, though yielding only modest quantitative gains on the validation set was essential for adapting the network to our specific sensor noise profile and environment textures.

Analysis of event camera noise characteristics reveals two primary components: random noise distributed across the sensor array and systematic noise from specific “hot pixels” that trigger events continuously. Under static conditions, 3767 pixels (1.22% of total) generate noise events over a 5-second observation period, with 99% of noise events concentrated in just 114 pixels (0.037% of total).

Hot pixel identification is performed through statistical analysis of event rates under static conditions. Events from identified hot pixels are filtered before network inference, significantly improving reconstruction quality with minimal information loss (Fig. 8). This statistical filtering step was a necessary methodological contribution, as it prevents the VIO algorithm from attempting to track false, static features generated by sensor noise, which would otherwise degrade pose estimation stability.

2.5. Visual-inertial odometry

The second stage processes reconstructed frames and IMU measurements using established visual-inertial odometry techniques. The system employs OpenVINS (Geneva et al., 2020), an open-source implementation of the Multi-State Constraint Kalman Filter (MSCKF) (Mourikis and Roumeliotis, 2007), chosen for its computational efficiency and real-time performance characteristics.

The MSCKF algorithm maintains a sliding window of camera poses and estimates 3D landmark positions through triangulation of tracked visual features across multiple frames. IMU measurements provide motion prediction and scale observability, addressing the inherent scale ambiguity of monocular visual odometry. The filter estimates the current UAV pose (position, orientation, velocity, and IMU biases) while marginalizing out older states to maintain computational tractability.

Fig. 9 illustrates the general VIO processing scheme and example trajectory estimation results, showing tracked landmarks (red points), current pose (coordinate frame), and estimated trajectory (blue line) in a 3D environment representation.

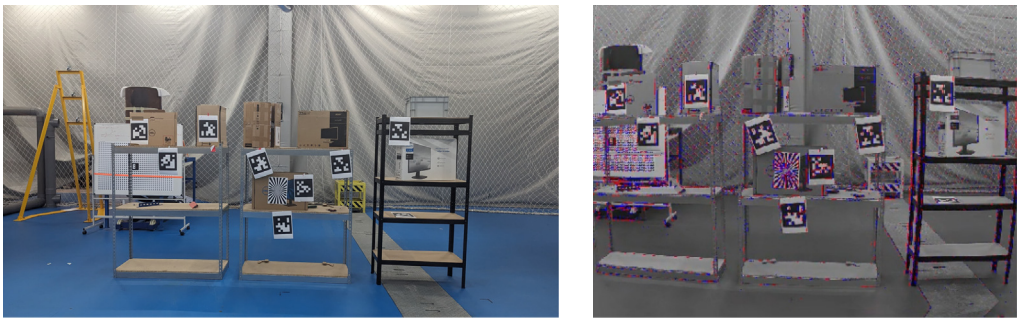


Fig. 7. Fine-tuning dataset generation: environment photograph (left) and synthetic event-frame pair (right).

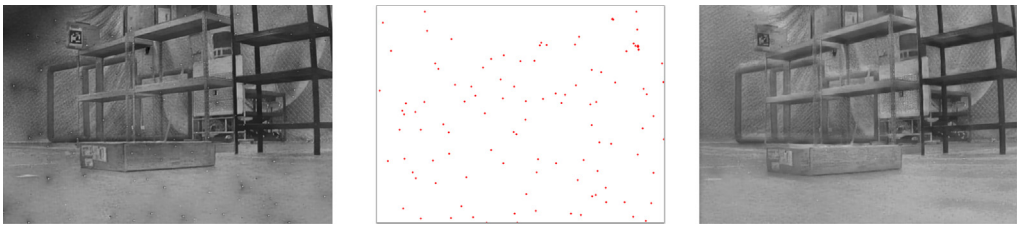


Fig. 8. Event camera noise mitigation: original reconstruction with noise artifacts (left), identified hot pixels (center), filtered reconstruction (right).

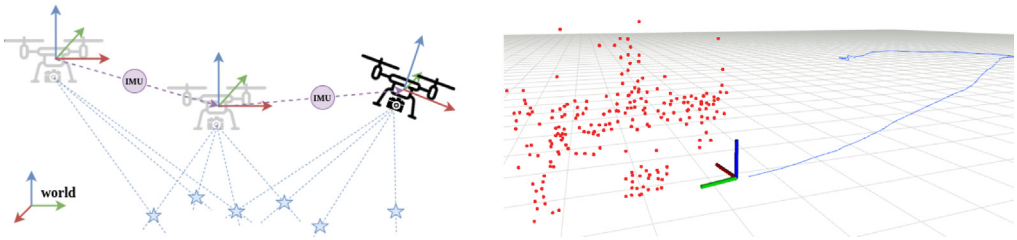


Fig. 9. Visual-inertial odometry implementation: algorithm overview (left) and trajectory estimation example (right).

2.6. Software implementation

The complete system is implemented within the Robot Operating System (ROS) framework running on Ubuntu 20.04, enabling modular development and real-time inter-process communication. The software architecture consists of three primary components:

- **Device interfaces**: ROS drivers for event camera and IMU data acquisition, with IMU measurements obtained directly from the UAV autopilot for consistency with flight control systems.
- **Frame reconstruction node**: Encapsulates the fine-tuned FireNet model for real-time inference at 30 Hz, generating reconstructed frames from accumulated event packages.
- **Visual-inertial odometry node**: OpenVINS implementation processing reconstructed frames and inertial measurements to estimate 6-DOF pose at rates up to 20 Hz.

Fig. 10 illustrates the complete software architecture, showing data flow between hardware interfaces (purple), processing nodes (green), and communication channels (blue). This modular design enables independent optimization of each component and facilitates future system extensions.

The implementation utilizes Python for neural network inference and data preprocessing, C++ for computationally intensive VIO processing, and PyTorch for deep learning operations. Three open-source software packages have been developed to support system deployment and future research.

A key engineering design choice in this architecture is the deliberate split of computational tasks. The `events_video_reconstructor` node (Fig. 10) is offloaded to the GPU to accelerate neural network inference. Simultaneously, the `openvins` node runs on the CPU for the filtering-based VIO. This parallel CPU/GPU pipeline is the core contribution that enables the system to achieve its real-time targets.

2.7. Performance considerations

The system is designed to operate within the computational and power constraints typical of small UAV platforms. Frame reconstruction achieves 30 Hz processing rates on the Jetson Xavier NX, while visual-inertial odometry maintains 20 Hz operation. Total system power consumption remains under 15 W, acceptable for multi-rotor platforms with flight times exceeding 15 min.

Event-based sensing is evolving quickly: HD-class sensors (~0.92 MP, 1280 × 720) such as Sony–Prophesee IMX636. Our VIO pipeline is resolution-agnostic and can scale with these sensors when power/compute budgets permit. Here we favored 640 × 480 to guarantee enough reconstruction and VIO frequencies within a 15 W envelope.

Processing latency from event detection to pose estimate is typically 50-80ms, meeting the requirements for UAV navigation control loops. The modular architecture enables optimization of individual components and adaptation to different hardware platforms with varying computational capabilities.

3. Experimental results

This section presents quantitative evaluation of the proposed event-based visual-inertial odometry system through controlled experiments

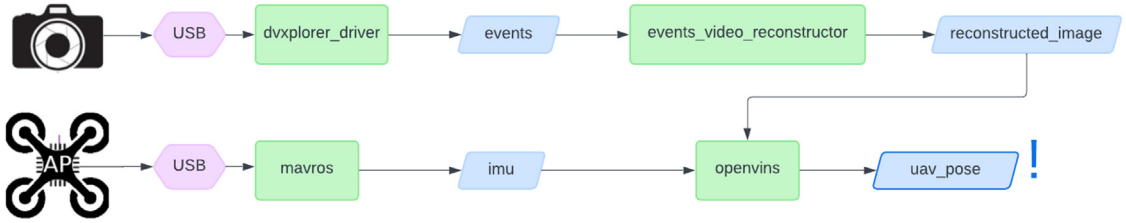


Fig. 10. Software architecture showing device interfaces, processing nodes, and communication channels.

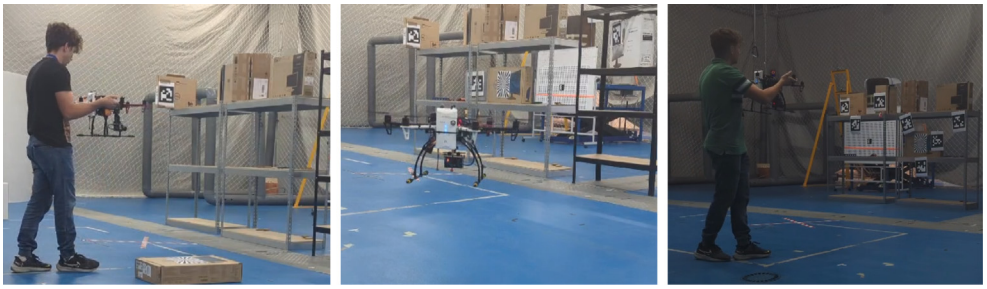


Fig. 11. Experimental scenarios: manual trajectory control (left), autonomous UAV flight (center), and dynamic lighting conditions (right).

in the CATEC multi-UAS laboratory. Three experimental scenarios evaluate different aspects of system performance: baseline accuracy under controlled conditions, performance during actual UAV flight, and operation under challenging lighting variations. All experiments utilize VICON motion capture data as ground-truth for pose estimation accuracy assessment.

3.1. Experimental design and metrics

Each experiment follows rectangular trajectory patterns consisting of three vertically stacked rectangles to facilitate observation of accumulated drift over extended operation periods. This trajectory design provides sufficient translational and rotational motion for robust odometry initialization while enabling systematic error analysis.

Performance evaluation employs standard metrics from visual odometry literature:

- **Absolute Pose Error (APE)**: Euclidean distance between estimated and ground-truth positions at each timestamp, indicating overall trajectory accuracy.
- **Relative Pose Error (RPE)**: Error in relative motion between consecutive poses, representing local consistency and drift characteristics.

Dynamic initialization is required for all experiments due to the event camera's dependence on motion for data generation. While this introduces initialization complexity, the OpenVINS framework handles this challenge effectively, though occasional minor initialization errors occur.

Fig. 11 illustrates the three experimental scenarios, with complete experimental footage available online[1] for comprehensive evaluation of system behavior.

3.2. Experiment 1: Baseline performance evaluation

The first experiment establishes baseline system performance under optimal conditions by manually guiding the UAV through the predetermined trajectory without flight-induced vibrations or control system dynamics.

[1] https://youtu.be/ouZyXximJoI

Table 2
Experiment 1 quantitative results under controlled conditions.

	Maximum	Mean	Standard Deviation
APE (m)	0.955	0.261	0.188
RPE (m)	0.129	0.005	0.006

The UAV is manually moved through three rectangular trajectories without yaw variations, eliminating vibrations and rapid dynamics that occur during powered flight. This controlled approach isolates the performance of the event-based odometry system from external disturbances.

Table 2 presents quantitative results showing mean absolute pose error of 26.1 cm with standard deviation of 18.8 cm. The relative pose error achieves significantly better performance with mean error of 4.9 mm and standard deviation of 6.5 mm, indicating good local consistency.

Fig. 14(a) shows trajectory estimation and error evolution over time. The largest absolute errors occur during initial trajectory segments, likely due to scale factor convergence inherent in monocular visual odometry systems. After initialization, the system maintains consistent performance with gradual drift accumulation typical of odometry approaches.

3.3. Experiment 2: Flight performance evaluation

The second experiment evaluates system performance during actual UAV flight, introducing realistic operational conditions including vibrations, aerodynamic disturbances, and flight control system dynamics.

A human pilot manually controls the UAV to follow the same rectangular trajectory pattern while minimizing altitude variations. This introduces realistic flight conditions including motor vibrations, aerodynamic effects, and faster motion dynamics compared to the baseline experiment.

Flight conditions result in increased error metrics compared to the baseline case (Table 3). Mean absolute pose error increases to 37.9 cm (45% increase), while relative pose error increases to 12.6 mm (158% increase). Despite these increases, the system maintains stable operation throughout the flight duration.

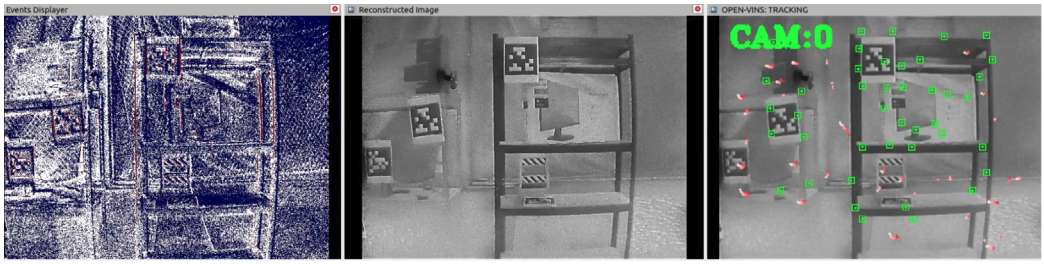


Fig. 12. Event camera response during illumination transition, showing negative polarity events (blue) while maintaining stable frame reconstruction. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 3
Experiment 2 quantitative results during UAV flight conditions.

	Maximum	Mean	Standard Deviation
APE (m)	1.169	0.379	0.213
RPE (m)	0.236	0.013	0.007

Table 4
Experiment 3 quantitative results under dynamic lighting conditions.

	Maximum	Mean	Standard Deviation
APE (m)	1.708	0.415	0.171
RPE (m)	1.428	0.012	0.030

The performance degradation is attributed to two primary factors: (1) high-frequency vibrations from UAV propulsion system affecting both event camera and IMU measurements, and (2) increased motion dynamics requiring faster algorithm response and feature tracking under more challenging conditions.

Fig. 14(b) demonstrates that despite increased error levels, the trajectory estimation maintains reasonable accuracy and the system does not experience catastrophic failures or divergence during flight operations.

3.4. Experiment 3: Dynamic lighting performance

The third experiment specifically evaluates the claimed advantage of event cameras under challenging lighting conditions by introducing rapid illumination changes during trajectory execution.

Returning to manual UAV guidance to isolate lighting effects from flight dynamics, the experiment introduces random lighting variations by switching room illumination on and off during trajectory execution. This tests the system's ability to maintain stable operation under the rapid lighting changes that typically cause failure in conventional camera systems.

Table 4 shows that despite challenging lighting conditions, the system maintains performance levels similar to baseline conditions. Mean absolute pose error (41.5 cm) remains comparable to flight conditions, while relative pose error (12.0 mm) stays consistent with previous experiments.

Fig. 12 captures a moment when room lighting is switched off, showing the event camera's response through numerous negative polarity events (blue) corresponding to the sudden intensity decrease. Importantly, the frame reconstruction process maintains visual quality and feature tracking remains stable throughout these transitions.

The error plots in Fig. 14(c) show a temporary spike corresponding to brief VICON tracking loss during one lighting transition, but the odometry system continues stable operation. This demonstrates the event camera's advantage in maintaining functionality under lighting conditions that would challenge conventional vision systems.

Table 5
Comparative performance analysis across experimental conditions.

Experiment	Mean APE (m)	Mean RPE (m)	Performance impact
Baseline (Manual)	0.261	0.005	Reference
Flight Conditions	0.379	0.013	+45% APE, +158% RPE
Dynamic Lighting	0.415	0.012	+59% APE, +140% RPE

3.5. Control: frame-based VIO under light toggles

A conventional frame camera is used under the same on/off light toggles scenario of Section 3.4. This camera consists of a RGB sensor (1280×720) from an Intel RealSense D450, processed using OpenVINS.

Fig. 13 shows the results of this experiment under a light variation. The frame-based pipeline shows a temporary collapse of all feature tracks at abrupt toggles, exposing a critical vulnerability. In the full experiment it recovered only thanks to the strong IMU fusion and the moderate motion of the test. The event-based pipeline shown in the previous Experiment 3, remains fully operational without exposure retuning or feature loss, as evidenced in Fig. 12.

3.6. Comparative analysis

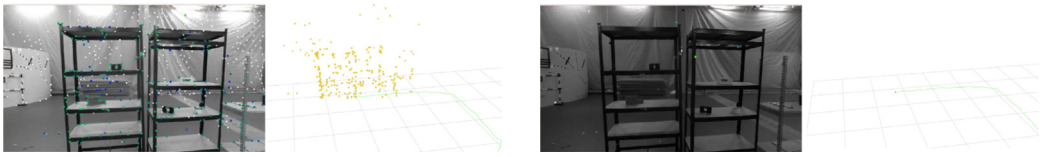
Table 5 summarizes performance metrics across all three experimental conditions, enabling direct comparison of system performance under different operational scenarios.

Key observations from the experimental results include:

- **Baseline accuracy**: The system achieves centimeter-level relative accuracy under controlled conditions, demonstrating the effectiveness of the event-based approach for indoor navigation.
- **Flight impact**: Real flight conditions increase error levels but maintain system stability, indicating the approach is viable for practical UAV applications.
- **Lighting resilience**: Dynamic lighting conditions do not significantly degrade performance compared to flight effects, validating the theoretical advantages of event cameras in challenging illumination scenarios.
- **Initialization sensitivity**: All experiments show increased errors during initial phases, suggesting opportunities for improved initialization strategies.

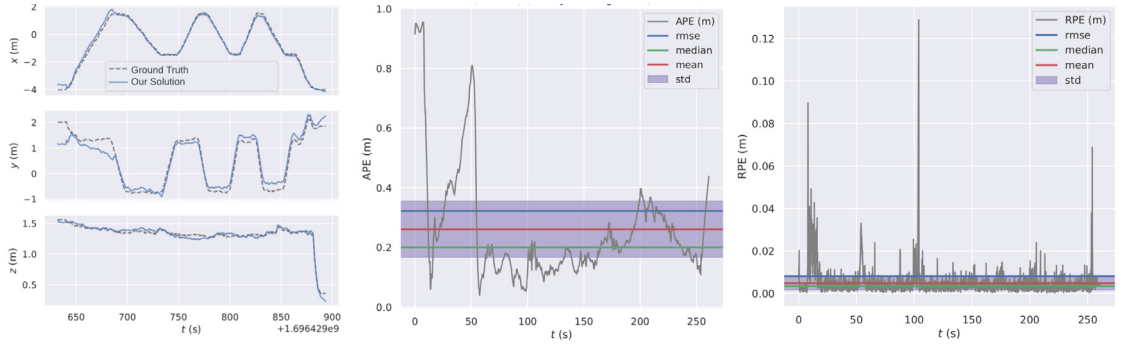
4. Discussion

This section analyzes the experimental results in the context of existing indoor navigation solutions, examines the practical implications of the proposed approach, and discusses limitations that inform future research directions.

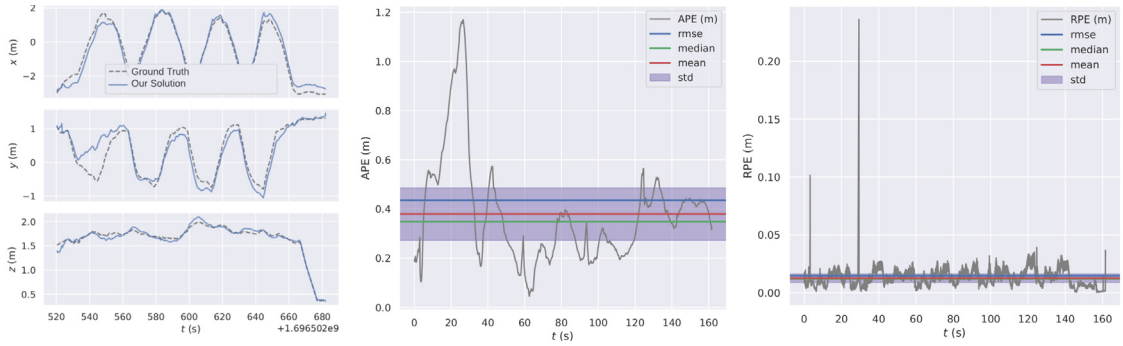


(a) Frame-based VIO (Lights ON) (b) Frame-based VIO (Lights OFF)

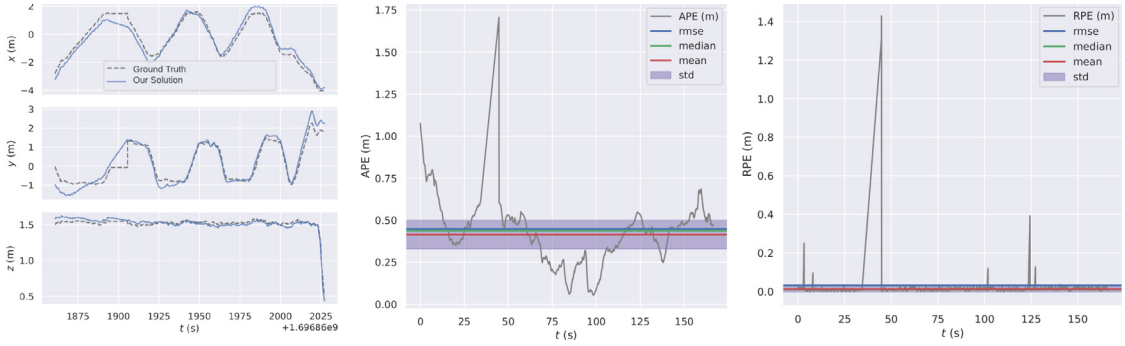
Fig. 13. Control experiment showing the frame-based VIO failure. (a) Stable feature tracking under normal illumination, just before the light toggle. (b) The same system, milliseconds later, showing an instantaneous and total loss of all features due to the under-exposed frame.



(a) Experiment 1: Baseline controlled conditions.



(b) Experiment 2: UAV flight conditions.



(c) Experiment 3: Dynamic lighting conditions.

Fig. 14. Experimental results showing estimated trajectory (left), absolute pose error (center), and relative pose error (right) for each test scenario.

4.1. Performance analysis and benchmarking

The experimental results demonstrate that event-based visual-inertial odometry achieves performance levels suitable for indoor UAV navigation applications. With mean absolute pose errors of 26–42 cm across different operational conditions, the system provides accuracy comparable to conventional visual-inertial systems operating under similar constraints. VINS-Mono typically achieves 0.1–0.5 m accuracy in structured indoor environments (Qin et al., 2018), while OpenVINS demonstrates similar performance ranges (Geneva et al., 2020). The proposed event-based system operates within this performance envelope while offering additional advantages under challenging conditions.

The relative pose error performance (5–13 mm mean) indicates good local consistency, suggesting that accumulated drift rather than instantaneous tracking errors represents the primary accuracy limitation. This characteristic is typical of visual odometry systems and can be

addressed through loop closure detection or external localization aids. The systematic performance degradation observed across experimental conditions provides insights into system behavior under realistic deployment scenarios. The 45% increase in absolute pose error during actual flight primarily results from vibration-induced measurement noise and increased motion dynamics that challenge feature tracking algorithms.

Importantly, the dynamic lighting experiment shows only modest additional performance degradation (59% increase) compared to flight conditions, validating the theoretical advantages of event cameras under challenging illumination. This represents a significant advantage over conventional camera systems, which typically experience complete tracking failure under rapid lighting transitions.

Moreover, under abrupt illumination changes, frame-based VIO can experience transient track collapse and later recover via IMU fusion, whereas the event-based pipeline avoids exposure retuning and motion-blur accumulation, providing steadier operation on embedded hardware.

4.2. Event camera advantages and practical limitations

The experimental results confirm several theoretical advantages of event cameras for UAV navigation. The system maintains stable operation during rapid illumination changes that would cause conventional visual-inertial systems to fail, enabling reliable navigation in environments with variable lighting conditions. The absence of frame-rate limitations enables continuous tracking without temporal gaps that affect conventional cameras during rapid motion. Additionally, the 30 Hz frame reconstruction rate combined with 20 Hz odometry processing demonstrates real-time capability suitable for UAV control loops while operating within power constraints typical of small aerial platforms.

However, several limitations emerged that inform future research directions. All experiments demonstrated increased errors during initial operation phases, reflecting the challenge of initializing monocular visual-inertial systems when motion is required for event generation. Flight conditions significantly increased error levels, suggesting that high-frequency vibrations affect both event generation and IMU measurements. Despite noise mitigation strategies, frame reconstruction occasionally produces artifacts that can affect feature tracking quality. The intermediate representation approach, while enabling compatibility with established algorithms, introduces an additional source of potential errors.

4.3. Implementation considerations and technology comparison

The system achieves real-time performance on the NVIDIA Jetson Xavier NX platform, consuming approximately 15 W total power. Frame reconstruction represents the most computationally intensive component, requiring GPU acceleration for 30 Hz operation. This power consumption is acceptable for multi-rotor platforms but may limit applicability to smaller fixed-wing UAVs with stricter power budgets. The modular software architecture enables optimization of individual components based on available computational resources.

While LiDAR systems typically achieve superior accuracy (sub-centimeter precision), the proposed event-based approach offers significant advantages in weight, power consumption, and cost. The 100–500 g weight penalty of 3D LiDAR systems substantially impacts UAV flight time, while the proposed system adds only 50 g to the overall platform weight. The DVXplorer camera (approximately $3000) combined with processing hardware represents a significant cost reduction compared to 3D LiDAR systems ($5000-$50,000), making the technology accessible for broader deployment scenarios.

Compared to conventional visual-inertial systems, the primary advantage lies in operational conditions where conventional cameras experience difficulties. Under nominal lighting and motion conditions, conventional systems may achieve slightly better accuracy due to mature algorithms and extensive optimization. However, the event-based system maintains consistent performance across a broader range of operational conditions.

4.4. Future research directions and limitations

Several algorithmic enhancements could improve system performance. Developing visual-inertial odometry algorithms that process events directly rather than through frame reconstruction could eliminate intermediate representation artifacts and improve computational efficiency. Dynamic adjustment of reconstruction parameters based on motion and lighting conditions could optimize frame quality for different operational scenarios. Integration of additional sensors such as conventional cameras for initialization or depth sensors for scale could address current limitations while maintaining event camera advantages.

Future work could explore loop closure detection using event-based descriptors to reduce accumulated drift in longer missions, extension to simultaneous localization and mapping capabilities, and application to multi-UAV collaborative navigation scenarios where event cameras' low latency advantages could prove beneficial.

Several factors limit the generalizability of the presented results. All experiments were conducted in a single indoor facility with known characteristics, and performance in different environments may vary significantly. The rectangular trajectory patterns may not represent the full range of motions encountered in practical UAV applications, and experiments spanning several minutes may not reveal long-term drift characteristics important for extended missions. Like all visual odometry systems, performance degrades in texture-poor environments, and event cameras may actually perform worse than conventional cameras in static, low-texture scenarios. Our future work plan will specifically target:

- Larger-scale environments to evaluate performance and drift over longer trajectories, beyond the current laboratory.
- Feature-poor and repetitive-texture scenes, such as long corridors, which are known failure cases for visual odometry.
- Complex and aggressive flight trajectories that go beyond simple rectangular patterns to better represent the complex maneuvers of real-world UAV missions.

4.5. Implications for indoor UAV navigation

The results demonstrate that event-based visual-inertial odometry represents a viable alternative to conventional approaches for indoor UAV navigation, particularly in challenging environmental conditions. The system achieves acceptable accuracy for applications such as inspection, monitoring, and autonomous navigation while offering advantages in lighting resilience and motion blur immunity. The cost-effectiveness and moderate computational requirements make the technology accessible for commercial and research applications where LiDAR systems would be prohibitive.

The initialization challenges and texture dependency suggest that hybrid approaches combining multiple sensing modalities may provide optimal solutions for demanding applications. For practical deployment, the system would benefit from integration with complementary technologies such as ultrawide-band positioning for absolute localization or conventional cameras for improved initialization. The modular architecture developed in this work facilitates such extensions while maintaining the core advantages of event-based sensing, and the open-source software contributions enable broader research community engagement and practical deployment.

5. Conclusions

This work demonstrates the feasibility of event-based visual-inertial odometry for indoor UAV navigation through a comprehensive system design and experimental validation. The proposed approach successfully addresses key limitations of conventional vision systems while maintaining computational efficiency suitable for embedded deployment.

The research makes several significant contributions to event-based navigation. The integration of event cameras with established visual-inertial odometry algorithms through intermediate frame reconstruction achieves mean absolute pose errors of 26–42 cm, comparable to conventional systems while offering enhanced performance under challenging conditions. The system maintains stable operation during rapid lighting transitions that cause conventional cameras to fail, with only 59% performance degradation compared to baseline conditions, validating the theoretical advantages of event-based sensing.

The comprehensive system implementation demonstrates real-time capability with 30 Hz frame reconstruction and 20 Hz odometry processing on embedded hardware, consuming 15 W total power and adding only 50 g to the UAV platform. The domain adaptation framework and noise mitigation strategies provide practical solutions for deploying event cameras in UAV applications, while the open-source software release enables broader community engagement and reproducible research.

Despite these achievements, several limitations require future investigation. The initialization complexity inherent in event-based systems and sensitivity to high-frequency vibrations during flight operations represent practical challenges for deployment. The intermediate reconstruction approach, while ensuring compatibility with existing algorithms, introduces computational overhead and potential artifacts that direct event processing could eliminate.

Future research should focus on developing algorithms that process events directly, implementing hybrid sensing approaches for improved initialization, and integrating loop closure capabilities to address long-term drift accumulation. Extended evaluation is required in larger-scale, feature-poor, and repetitive-texture environments, using aggressive flight trajectories to fully establish the technology's practical applicability beyond the baseline presented in this work.

The results establish event-based visual-inertial odometry as a viable alternative for indoor UAV navigation, particularly in applications where environmental robustness is prioritized over marginal accuracy improvements. The cost-effectiveness relative to LiDAR systems and consistent performance across varying operational conditions make this technology accessible for commercial applications and research scenarios where premium sensing solutions are not feasible.

This work provides essential groundwork for the practical deployment of event-based navigation systems, offering both technical solutions and implementation insights. As event camera technology continues to mature, the modular architecture and systematic evaluation methodology presented here will support the development of more sophisticated navigation systems that fully exploit the unique advantages of neuromorphic sensing for autonomous aerial systems.

CRediT authorship contribution statement

David Tejero-Ruiz: Software, Methodology, Formal analysis, Data curation, Conceptualization. **David Solís-Martín:** Writing – review & editing, Writing – original draft, Validation, Supervision, Project administration, Funding acquisition. **Francisco J. Pérez-Grau:** Writing – review & editing, Writing – original draft, Validation, Project administration. **Joaquín Borrego-Díaz:** Writing – review & editing, Validation, Supervision, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

Grant PID2023-147198NB-I00 funded by MICIU/AEI/10.13039/501100011033 (Agencia Estatal de Investigación), Spain and by FEDER, UE, Spain.

Data availability

No data was used for the research described in the article.

References

Caballero, R., Parra, J., Trujillo, M.Á., Pérez-Grau, F.J., Viguria, A., Ollero, A., 2021. Aerial robotic solution for detailed inspection of viaducts. Appl. Sci. 11 (18), 8404.

Cook, M., Gugelmann, L., Jug, F., Krautz, C., Steger, A., 2011. Interacting maps for fast visual interpretation. In: The 2011 International Joint Conference on Neural Networks. IEEE, pp. 770–776.

Delbruck, T., Lang, M., 2013. Robotic goalie with 3ms reaction time at 4% CPU load using event-based dynamic vision sensor. Front. Neurosci. 7, 223. http://dx.doi.org/10.3389/fnins.2013.00223.

Ercan, B., Eker, O., Saglam, C., Erdem, A., Erdem, E., 2024. Hypere2vid: Improving event-based video reconstruction via hypernetworks. IEEE Trans. Image Process.

Falanga, D., Kleber, K., Scaramuzza, D., 2020. Dynamic obstacle avoidance for quadrotors with event cameras. Sci. Robot. 5 (40), eaaz9712.

Gallego, G., Lund, J.E., Mueggler, E., Rebecq, H., Delbruck, T., Scaramuzza, D., 2018. Event-based, 6-DOF camera tracking from photometric depth maps. IEEE Trans. Pattern Anal. Mach. Intell. 40 (10), 2402–2412. http://dx.doi.org/10.1109/TPAMI.2017.2769655.

Geneva, P., Eckenhoff, K., Lee, W., Yang, Y., Huang, G., 2020. Openvins: A research platform for visual-inertial estimation. In: 2020 IEEE International Conference on Robotics and Automation. ICRA, pp. 4666–4672. http://dx.doi.org/10.1109/ICRA40945.2020.9196524.

Glover, A., Bartolozzi, C., 2017. Robust visual tracking with a freely-moving event camera. In: 2017 IEEE/RSJ International Conference on Intelligent Robots and Systems. IROS, pp. 3769–3776. http://dx.doi.org/10.1109/IROS.2017.8206226.

Hidalgo-Carrió, J., Gallego, G., Scaramuzza, D., 2022. Event-aided direct sparse odometry. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition. pp. 5781–5790.

Macario Barros, A., Michel, M., Moline, Y., Corre, G., Carrel, F., 2022. A comprehensive survey of visual slam algorithms. Robotics 11 (1), 24.

Minoda, K., Schilling, F., Wüest, V., Floreano, D., Yairi, T., 2021. Viode: A simulated dataset to address the challenges of visual-inertial odometry in dynamic environments. IEEE Robot. Autom. Lett. 6 (2), 1343–1350.

Mourikis, A.I., Roumeliotis, S.I., 2007. A multi-state constraint Kalman filter for vision-aided inertial navigation. In: Proceedings 2007 IEEE International Conference on Robotics and Automation. pp. 3565–3572. http://dx.doi.org/10.1109/ROBOT.2007.364024.

Patrik, A., Utama, G., Gunawan, A.A.S., Chowanda, A., Suroso, J.S., Shofiyanti, R., Budiharto, W., 2019. GNSS-based navigation systems of autonomous drone for delivering items. J. Big Data 6, 1–14.

Qin, T., Li, P., Shen, S., 2018. Vins-mono: A robust and versatile monocular visual-inertial state estimator. IEEE Trans. Robot. 34 (4), 1004–1020.

Rebecq, H., Gehrig, D., Scaramuzza, D., 2018. ESIM: an open event camera simulator. In: Conf. on Robotics Learning. CoR).

Rebecq, H., Ranftl, R., Koltun, V., Scaramuzza, D., 2019. High speed and high dynamic range video with an event camera. IEEE Trans. Pattern Anal. Mach. Intell. 43 (6), 1964–1980.

Scheerlinck, C., Rebecq, H., Gehrig, D., Barnes, N., Mahony, R., Scaramuzza, D., 2020. Fast image reconstruction with an event camera. In: Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision. pp. 156–163.

Seiskari, O., Rantalankila, P., Kannala, J., Ylilammi, J., Rahtu, E., Solin, A., 2022. Hybvio: Pushing the limits of real-time visual-inertial odometry. In: Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision. pp. 701–710.

Shakhatreh, H., Sawalmeh, A.H., Al-Fuqaha, A., Dou, Z., Almaita, E., Khalil, I., Othman, N.S., Khreishah, A., Guizani, M., 2019. Unmanned aerial vehicles (UAVs): A survey on civil applications and key research challenges. IEEE Access 7, 48572–48634.

Sun, J., Zhao, J., Hu, X., Gao, H., Yu, J., 2023. Autonomous navigation system of indoor mobile robots using 2D lidar. Mathematics 11 (6), http://dx.doi.org/10.3390/math11061455.

Vidal, A.R., Rebecq, H., Horstschaefer, T., Scaramuzza, D., 2018. Ultimate SLAM? Combining events, images, and IMU for robust visual SLAM in HDR and high-speed scenarios. IEEE Robot. Autom. Lett. 3 (2), 994–1001. http://dx.doi.org/10.1109/LRA.2018.2793357.

Vitale, A., Renner, A., Nauer, C., Scaramuzza, D., Sandamirskaya, Y., 2021. Event-driven vision and control for UAVs on a neuromorphic chip. In: 2021 IEEE International Conference on Robotics and Automation. ICRA, IEEE, pp. 103–109.

Weng, W., Zhang, Y., Xiong, Z., 2021. Event-based video reconstruction using transformer. In: Proceedings of the IEEE/CVF International Conference on Computer Vision. pp. 2563–2572.

Zhang, X., Tie, J., Li, J., Hu, Y., Liu, S., Li, X., Li, Z., Yu, X., Zhao, J., Wan, Z., et al., 2023. Dynamic obstacle avoidance for unmanned aerial vehicle using dynamic vision sensor. In: International Conference on Artificial Neural Networks. Springer, pp. 161–173.

Zhou, Y., Gallego, G., Shen, S., 2021. Event-based stereo visual odometry. IEEE Trans. Robot. 37 (5), 1433–1450.